Project Proposal: Agentic Health Coach

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Abstract

Wearable devices can continuously capture rich physiological and behavioral data, yet most health technologies still require users to interpret data or initiate interactions on their own. This creates friction between sensing and action, limiting the usefulness of wearable health data in everyday life. Recent large language model (LLM) systems show promise for making personal health data more understandable, while prior work on proactive interventions highlights the importance of timing, restraint, and user control.

We propose an agentic health coach that uses Apple Watch data to generate proactive, low-friction health recommendations through mobile and wearable interactions. The system combines LLM-based health sensemaking with context-aware intervention delivery, aiming to provide timely support without relying on either reactive chat-based interaction or static rule-based notifications. To study this design, we plan a prototype-based user study examining how intervention timing, explanation style, tone, and privacy controls shape perceived usefulness, trust, and acceptance. This work aims to contribute design insights for proactive wearable health agents and broader HCI discussions around personal informatics, health AI, and agentic interactive systems.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; *Natural language interfaces*; • **Applied computing** → *Health informatics.*

Keywords: wearable computing, large language models, health coaching, proactive interventions, mobile health, trust and privacy

1 Introduction

Wearable devices such as the Apple Watch can continuously capture rich physiological and behavioral signals, including activity, heart rate, and sleep. In principle, these data streams could support healthier daily decisions and more personalized behavior change. In practice, however, most health technologies still place the burden of interpretation and action on the user. People are expected to open an app, inspect dashboards, interpret unfamiliar metrics, and decide what to do next. As a result, there remains a substantial gap between passive sensing and meaningful behavior change [2, 10, 16].

This gap matters because behavior change is not driven by information alone. Prior work in psychology and digital health has shown that effective support depends on whether an intervention is delivered at an appropriate moment, in an understandable form, and in a way that aligns with a user's goals and readiness to act. Just-In-Time Adaptive Interventions (JITAIs) provide one important framework for this problem: rather than delivering static reminders on fixed schedules, JITAIs aim to provide support when a person is most likely to benefit, based on changing context, internal state, and receptivity [7, 13]. However, many existing JITAI-style systems still rely on hand-authored messages, narrow intervention types, or relatively rigid decision rules, which can produce repetitive experiences and eventual disengagement [1, 14].

At the same time, recent work on large language models (LLMs) suggests a new opportunity for personal health support. Systems such as PhysioLLM, GPTCoach, HealthGURU, and the Personal Health Insights Agent have shown that LLMs can help translate complex wearable or health-tracking data into more understandable, personalized, and actionable insights [2, 5, 10, 20]. These systems begin to address a core limitation of commercial health apps: users often need explanation and guidance, not just measurement. Yet most LLM-based health systems remain reactive. They typically require users to initiate a conversation, ask a question, or navigate to a phone-based chatbot, which introduces friction precisely when support should be easy and immediate [5, 10].

Research on notification timing and delivery further suggests that proactive health support is only effective under specific conditions. Prior studies show that contextual notifications outperform simple time-based reminders in many cases, that receptivity models can substantially improve the likelihood that users engage with an intervention, and that smartwatch delivery modalities such as combined haptic and visual feedback can improve response and compliance

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[1, 3, 11, 15]. At the same time, poorly timed or repetitive notifications can reduce trust, create alert fatigue, and contribute to disengagement [7, 14]. This literature suggests that the challenge is not merely whether to be proactive, but how to be proactive without becoming intrusive.

A further challenge is that health support systems must operate within strong boundaries of trust and privacy. Studies on health AI, remote patient monitoring, and LLM-based interfaces show that users may disclose sensitive health information to AI systems when those systems feel nonjudgmental or helpful, while simultaneously expressing concern about data retention, opacity, and inappropriate secondary use [8, 9, 18]. Trust is shaped not only by accuracy, but also by explanation quality, usability, transparency, and interface form [17, 19]. For a proactive wearable-based health assistant, this means that success depends not only on generating good recommendations, but on doing so in ways that users perceive as understandable, bounded, and respectful of their control over personal data [4, 17].

Motivated by these gaps, we propose an agentic health coach that continuously interprets Apple Watch data and delivers proactive, real-time recommendations through low-friction wearable interactions. Our project sits at the intersection of three lines of prior work: wearable and mobile interventions for health behavior change, LLM-based health sensemaking, and research on notification timing, privacy, and trust. Unlike existing reactive health chatbots, our system is designed to monitor data streams in the background and surface concise, context-aware recommendations when intervention may be most useful. Unlike traditional proactive systems that rely on fixed or repetitive messaging, it uses an LLM to generate more adaptive explanations and support. Our goal is not simply to make health recommendations more intelligent, but to understand how proactivity, timing, explanation quality, communication tone, and privacy controls shape users' perceptions of usefulness, trust, and acceptance in an everyday wearable health assistant.

2 Related Work

Prior work relevant to our project falls into four themes: wearable health technologies and behavior change, large language model (LLM) systems for health sensemaking, proactive intervention timing and notification design, and trust and privacy in health AI.

2.1 Wearable Health Technologies and Behavior Change

Wearable health technologies have made continuous self-tracking more accessible, but prior research suggests that sensing alone does not guarantee meaningful behavior change. Reviews of fitness technologies show that wearables can support awareness and motivation, yet effects often depend on how feedback is framed, whether it aligns with user goals, and how much effort is required to interpret the data [16]. These findings align with behavior change perspectives such as the Health Belief Model and the Transtheoretical Model, which emphasize perceived benefits, barriers, readiness, and cues to action. Our project builds on this literature by treating wearable sensing as the input to a coaching system rather than the endpoint of the interaction.

2.2 LLMs for Health Sensemaking and Coaching

Recent work shows that LLMs can help people interpret complex personal health data. PhysioLLM translates wearable signals into personalized insights, while Vital Insight supports experts' context-driven sensemaking of multi-modal tracking data through visualization and human-in-the-loop LLM agents [2, 6]. Other systems such as GPTCoach, data-driven theory-guided sleep coaching, and LLM-agent pipelines for personal health insights demonstrate the promise of generating tailored advice from behavioral and physiological data [5, 10, 20]. More broadly, prior CHI work on conversational AI for public health interventions highlights both the opportunity and the challenge of using LLMs in health contexts: natural language can make support feel more accessible and adaptive, but systems must still be safe, understandable, and appropriate for sustained use [4]. Our project builds on these advances in health sensemaking, but differs from much of this literature by focusing on proactive wearable delivery rather than primarily reactive or chat-initiated interactions.

2.3 Proactive Timing, Receptivity, and Notification Modality

Another major line of work examines when and how digital health systems should intervene. The Just-in-Time Adaptive Intervention (JITAI) literature provides the conceptual basis for delivering support when a user is most likely to benefit in context rather than on a fixed schedule [13]. Work on treatment timing argues that interventions must be not only timely but also limited in frequency, since excessive prompting can reduce effectiveness [7]. Subsequent studies show that disengagement is a serious risk in mobile health, and that receptivity varies with daily context, activity, and cognitive state [11, 12, 14]. More recent work refines this picture: contextual notifications can improve self-monitoring in weight maintenance apps, there appears to be a "Goldilocks" time window for wearable AI interventions, and smartwatch notification modality can change responsiveness in sedentary behavior interventions [1, 3, 15]. Prior work also suggests that long-term acceptance depends on systems remaining sensitive to burden, interruptiveness, and user control rather than maximizing the number of prompts [7, 14]. Our project extends this literature by combining proactive timing with richer generated content and by examining how timing and explanation interact in users' perceptions of helpfulness.

2.4 Trust, Disclosure, and Privacy in Health AI

Health AI systems must also address whether people trust the system enough to rely on it and disclose the data it needs. Research on remote patient monitoring shows that willingness to share information with AI depends on expected benefit, perceived judgment, and the role of the advisor [9]. Studies of LLM-based chatbots and custom GPT-like systems similarly show that privacy concerns are shaped by perceived purpose, expectations about data handling, and uncertainty about who can access stored conversations [8, 18]. In parallel, work on interface trust suggests that interface form and explanation quality shape whether users believe health information from LLM systems [17]. Related evidence from reviews of trust in AI-based clinical decision support systems further suggests that transparency, usability, and integration into existing routines are central to long-term trust [19]. These findings are especially important for our project because a proactive wearable coach may feel more intrusive than a reactive chatbot if it acts too often or uses sensitive data without clear justification. Therefore, our design explicitly treats trust, explanation, and user control as core requirements rather than secondary interface concerns.

Across these themes, the main gap is clear. Existing LLM-based health systems are increasingly good at interpreting personal data, but many still depend on high-friction, user-initiated interactions. Existing proactive health systems are increasingly good at timing, but they often rely on static rules, narrow messages, or limited explanation. Our project addresses this gap by exploring an agentic health coach that combines wearable sensing, LLM-based interpretation, and context-aware delivery in a single system while also foregrounding timing, transparency, tone, privacy, and user control.

3 Methodology

We plan a mixed-method, within-subjects prototype study to evaluate the proposed health coach. Rather than testing a fully deployed system in the wild, this study uses a scenario-based prototype walkthrough to examine how users respond to proactive wearable health support at the level of interaction design. This approach is appropriate for an early-stage class project because it allows us to study the key design questions raised by prior work—namely, whether proactive delivery reduces friction relative to reactive interfaces, how timing and explanation shape trust and receptiveness, and what privacy and control boundaries users expect from a system that continuously interprets wearable data. The study is designed to evaluate not only whether the prototype seems useful in principle, but whether it feels appropriately timed, understandable, non-intrusive, and acceptable for everyday health support.

3.1 Research Questions and Hypotheses

The study is guided by three research questions:

RQ1: How do users perceive a proactive wearable health coach that combines LLM-based interpretation with context-aware delivery, compared with more reactive health interfaces?

RQ2: How do timing, explanation, and communication style shape the perceived helpfulness, trustworthiness, and intrusiveness of recommendations from a proactive wearable health coach?

RQ3: What forms of privacy control and user control do people expect from a proactive wearable health coach that continuously interprets wearable data?

Because this is an initial prototype study, our hypotheses focus only on the factors that we can directly manipulate in the prototype:

H1: Participants will perceive the proactive wearable health coach as more useful and less effortful than a reactive, chat-based health interface.

H2: Recommendations that include concise explanations will be perceived as more trustworthy and more helpful than recommendations without explanation.

H3: Recommendations delivered at contextually appropriate times will be perceived as less intrusive and more actionable than recommendations delivered at fixed times.

Questions about privacy preferences, acceptable automation, and preferred communication tone will be studied more exploratorily through participant judgments and follow-up interviews rather than treated as strong causal claims.

3.2 Participants

Our target population is college students and other young adults who already use, or are open to using, wearable devices such as the Apple Watch to monitor aspects of their daily health and routines. This population is appropriate because these users regularly navigate sleep, exercise, stress, deadlines, and competing routines, making them a good match for studying the value and burden of proactive health support in everyday life.

For an initial prototype study, we plan to recruit approximately 5–10 participants through university mailing lists, course forums, campus organizations, and word of mouth. Participants will be screened to ensure that they are at least 18 years old, regularly use a smartphone, and have experience with or interest in wearable-supported health tracking. Although the sample size is modest, it is appropriate for a formative HCI evaluation focused on identifying design preferences, friction points, and boundary conditions rather than establishing broad generalizability.

3.3 Study Design

The study will use a within-subjects design in which each participant experiences multiple versions of the prototype

across a set of standardized health scenarios. This allows us to compare interface styles and recommendation designs while keeping the underlying health context constant.

Participants will first complete a short intake questionnaire about their current health goals, routines, prior experience with wearable devices, and attitudes toward AI-supported health feedback. They will then be introduced to the prototype and guided through three study blocks.

In the first block, participants will compare a reactive baseline with the proactive wearable prototype. In the reactive condition, participants must explicitly open the app and request an explanation or recommendation. In the proactive condition, the system surfaces a recommendation automatically through a smartwatch-style notification. Both conditions will use the same underlying health scenario and recommendation content so that the main difference is the style of interaction rather than the advice itself. This block is designed to address RQ1 by examining whether proactive delivery feels more convenient and lower-friction than a reactive interface.

In the second block, participants will evaluate different recommendation variants within the proactive prototype. These variants will manipulate factors such as notification timing, explanation, and communication style. For timing, participants will compare recommendations delivered according to a static schedule with recommendations delivered in a context-aware manner based on the user's recent state or activity. For example, a recommendation might appear either at a fixed scheduled time or after a relevant trigger such as poor sleep, prolonged inactivity during study hours, or a stressful period during a busy day. Recommendations may also include either a concise explanation or no explanation, and may be written in a more neutral versus more supportive tone. These conditions will be presented through realistic health scenarios. This block is designed to address RQ2 by examining how timing strategy and presentation style shape whether proactive support feels helpful, trustworthy, or intrusive.

In the third block, participants will evaluate privacy and control scenarios for the proactive health coach. They will be shown settings screens, feature descriptions, or short vignettes that vary what data the system uses, how recommendations are justified, and what controls are available to the user. Example questions may include whether the system should be allowed to use sleep, activity, location, or stress-related signals; whether users should be able to mute certain recommendation types; whether the system should explain why a recommendation appeared; and what forms of data storage or personalization feel acceptable. This block is designed to address RQ3 by identifying what boundaries and controls participants consider necessary for acceptable use.

Throughout the session, participants will be encouraged to think aloud as they react to the prototype. After completing the three blocks, we will conduct a short semi-structured interview to probe what made the system feel helpful, disruptive, credible, too invasive, too robotic, or appropriately supportive.

3.4 Measurements and Evaluation Plan

We will collect both quantitative and qualitative data. Quantitative data will include pre-study background responses and post-task ratings for each condition. After each task block, participants will rate constructs such as perceived usefulness, trust, effort, intrusiveness, timing appropriateness, explanation clarity, privacy comfort, and willingness to act on the recommendation. We may also use the System Usability Scale (SUS) for overall usability, although the primary emphasis of the study is less on generic usability and more on perceived fit, trust, and acceptability of proactive health support.

Qualitative data will come from think-aloud comments and follow-up interviews. These data will help us understand not only which conditions participants preferred, but why. In particular, we will look for recurring themes around appropriate proactivity, preferred explanation style, acceptable communication tone, privacy boundaries, and desired user controls. We will analyze these responses using thematic analysis.

Because this is a small formative study, quantitative results will primarily be summarized descriptively through means, preference patterns, and condition comparisons rather than treated as definitive statistical evidence. If the final number of participants and study structure support simple paired comparisons, we may report those as exploratory analyses. Overall, the goal of this evaluation is to surface design insights about how a proactive wearable health coach should communicate, intervene, and respect user boundaries, rather than to claim final proof of effectiveness in real-world deployment.

4 Prototype Development

The prototype is designed as a mobile-first health coaching system centered on an API-based agent that can interpret wearable data and deliver timely recommendations without requiring constant manual user input. Rather than running the full agent locally on the device, the current design uses a remote model accessed through an API. This architecture makes it easier to iterate on prompting, tool use, reasoning pipelines, and future model updates while still allowing the mobile application to act as the primary user-facing interface.

The system will connect the mobile app to backend agent services that can read Apple Watch-derived health data, schedule periodic check-ins, and trigger notifications to the

user. At configurable times throughout the day, the application will retrieve recent wearable signals and send the relevant context to the API-based agent, which will determine whether a notification or health insight should be delivered. Users will also be able to define custom times of interest, such as workout periods, so the system can provide more context-aware support before, during, or after those activities. In addition, the app may support explicit reminders that can either be simple prompts or agent-generated suggestions.

During onboarding, the user will specify personal health goals, such as improving sleep, increasing activity, maintaining workout consistency, or developing healthier eating habits. The agent will use these goals together with incoming wearable data to estimate progress, identify potential obstacles, and generate personalized suggestions intended to improve the user's likelihood of meeting those goals. The system will also support optional meal tracking so that nutrition data can be incorporated into the agent's reasoning when relevant.

To personalize feedback, the prototype will collect baseline demographic and body-related information such as age, height, weight, and gender. These inputs can help the system interpret health signals within a more appropriate range for the individual user. If the core agent knowledge is not sufficient, the backend may include an additional retrieval or tool layer that returns normative or goal-relevant health ranges based on user demographics, allowing the agent to better contextualize its recommendations.

Because this prototype relies on an API-based agent, an important part of the system design is deciding what data is transmitted, how often requests are made, and how responses are cached or filtered before being surfaced to the user. This architecture also creates important privacy and reliability tradeoffs that will be considered in the system design, especially since health-related data is sensitive and recommendations should remain timely and understandable.

Finally, the prototype may allow limited customization of the agent's communication style or personality. This feature would support our broader research interest in understanding whether tone, presentation style, and perceived personality influence trust, motivation, and user acceptance of proactive health recommendations.

5 Expected Outcomes

After thinking through this project, one key outcome we expect is that an agentic health coach will feel more useful than traditional health apps that are reactive in the same way, but only when it reduces effort without becoming overwhelming. Overall the main advantage of this system is that it can interpret wearable data and giverecommendations without the user having to constantly check or request information. Threfore, the participants will likely see it as more convenient and easier to use in everyday situations where they are busy or not actively thinking about their health.

At the same time, another outcome we expect is that proactivity by itself will not lead to better experiences. For instance if the system interrupts at the wrong time or gives suggestions without reasoning, users could likely see it as intrusive rather than being something that helps them long term. This means timing and explanation will play a major role in how the system is perceived. Participants will probably be more willing to accept recommendations when they understand why they are being given and what data the system is using. Without that, even good recommendations in that aspect may feel random or unnecessary.

Another key takeaway we expect is that users will not judge the system based only on usefulness. Rather, they will evaluate it based on a balance between helpfulness, privacy, and control. For example, participants may be comfortable receiving proactive suggestions about general behaviors like sleep or activity, but less comfortable with more sensitive areas like diet or stress. This suggests that acceptable levels of automation depend on context rather than being fixed across all features. So even if the system works well technically, it still needs to respect different boundaries depending on the type of data and behavior involved.

We also expect that users will prefer systems that give them control over how the agent behaves. Participants will likely respond better when they can adjust things like notification frequency, timing, and the type of recommendations they receive. This reflects a core tension in the design of agentic systems. If the system does too little, it is not useful. If it does too much, it becomes annoying or uncomfortable. Because of that, giving users the ability to tune the system themselves will likely be an important factor in acceptance.

Another important outcome we expect is that communication style will affect how users interpret the system. Recommendations that feel clear and supportive will likely be received better than ones that feel overly robotic. So in a health context especially, tone really matters because users can already feel pressure around behavior change. So given two recommendations that are identical in content, the way they are presented is going to change whether they are taken seriously.

From a usability perspective, we would expect participants using the agentic prototype to report overall lower effort and higher usefulness compared to a baseline. This would support the idea that proactive systems can reduce the burden of managing health routines. Many current tools depend on users taking initiative, which creates inconsistency. Shifting some of that responsibility to the system may improve engagement, as long as the system remains predictable and understandable.

If we look at the qualitative data, we expect to see mixed reactions that help define where the system works and where

it fails. Some participants will likely describe it as helpful and motivating, while others may feel that it is too persistent or too aware of their behavior. This contrast is important because it shows that the goal is not just to make the system more intelligent, but to make it feel appropriate in how it acts.

Overall, the main outcome we expect is not just that an agentic health coach is useful, but that its success depends on how well it balances proactivity with user control. The project should help clarify how systems like this should deliver recommendations, explain their reasoning, and respect user boundaries.

6 References

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